Migrations and schema

English version of ../../database/migrations.md.

Tool

Alembic, with a single initial revision backend/alembic/versions/e2df629b67e6_initial_schema.py. Configuration — alembic.ini, environment — backend/alembic/env.py.

Commands

```
make alembic-upgrade DATABASE_URL='postgresql+asyncpg://user:pass@host:5432/db'
make alembic-current DATABASE_URL='...'
make alembic-check        # model/migration consistency (tests/test_schema_sync.py)
```

Locally SQLite (aiosqlite) is used by default, so tests and development need no external database. Production is PostgreSQL 16.

Rules

1. **One revision = one intention.** The message reflects the schema change, not a ticket number.
2. **Never edit an applied revision.** If a fix is needed, add a new revision.
3. **Models and migrations must match.** tests/test_schema_sync.py compares SQLAlchemy metadata with the history; a mismatch fails CI.
4. **Data is separate from schema.** Demo content is scripts/seed.py, not a migration.
5. **SQLite-only changes are not acceptable** as the only option: production is PostgreSQL.

Upgrading an installed instance

1. Backup (pg_dump; for files — the S3/MinIO bucket).
2. make alembic-upgrade on the new image version.
3. make alembic-current — the version must match head.
4. make smoke-test — /healthz, /readyz, the frontend.
5. Check one flow by hand: board, card, report.

This is verified automatically in .github/workflows/portability.yml: installation from scratch, a repeated upgrade head, the backup → restore → smoke cycle.